2027 BATCH · DETAILED SYLLABUS

TCS NQT
Topic-wise Syllabus

Every section broken into topics and sub-topics, with a priority level for each, for NQT Foundation and Advanced.

01 · NUMERICAL ABILITY	15 topics
02 · REASONING ABILITY	13 topics
03 · VERBAL ABILITY	5 topics
04 · ADVANCED QUANTITATIVE + REASONING	10 topics
05 · ADVANCED CODING	16 topics

01 NUMERICAL ABILITY

Part A · Foundation

20 Qs · 25 min · ~75 sec each · Moderate

Calculation speed and formula recall decide this section. Expect word problems, some TITA (typed) answers, and mixed-level questions.

Priority is our preparation guide based on how often the topic is practised for NQT: **High** · **Medium** · **Low**. TCS does not publish an official topic weightage.

Number System — High

- Types of numbers, divisibility rules
- Unit digit and last two digits of powers
- Remainders, remainder theorem
- Factors, number of factors, sum of factors
- Prime numbers, co-primes
- Base conversion (binary, octal, hex)
- Factorial and trailing zeros
- Recurring decimals, surds and indices

LCM & HCF — High

- Prime factorisation method
- Product of two numbers = LCM x HCF
- LCM/HCF of fractions
- Bells ringing together, circular track meeting
- Greatest number dividing with given remainders
- Smallest number divisible by a set of numbers

Percentages — High

- Percentage to fraction conversions
- Successive percentage change
- Population increase/decrease
- Percentage in marks, votes, salary problems
- Consumption vs price change
- Net change and expenditure problems

Profit & Loss — High

- CP, SP, MP, markup and discount
- Successive discounts
- Dishonest dealer (false weights)
- Profit on whole vs part
- Partial sale at different profit/loss
- Article sold at same SP with gain and loss

Time, Speed & Distance — High

- Average speed, relative speed
- Trains: crossing poles, platforms, trains
- Boats and streams (upstream/downstream)
- Races and circular tracks
- Meeting and chasing problems
- Late/early arrival problems

Time & Work — High

- Individual and combined work rate
- Efficiency and total work method
- Pipes and cisterns, leakage
- Alternate-day work
- Work and wages
- Men-days-hours (chain rule)

Simple & Compound Interest — Medium

- SI formula, amount and time
- CI annual, half-yearly, quarterly
- Difference between CI and SI
- Population growth as CI
- Installments (equal EMI)
- Interest rate and time unknown

Averages — Medium

- Average of consecutive numbers
- Weighted average
- Change in average with addition/removal
- Age-based averages
- Average speed and temperature
- Replacement of members

Ratio, Proportion & Partnership — Medium

- Ratio simplification, compound ratio
- Direct and inverse proportion
- Mean proportional, third proportional
- Division of amount in ratio
- Partnership with time and investment
- Variation (direct, inverse, joint)

Probability — Medium

- Coins, dice, cards
- Mutually exclusive and independent events
- At least / at most conditions
- Balls drawn from a bag
- Conditional probability (basic)
- Odds in favour and against

Permutation & Combination

Medium

- Fundamental counting principle
- nPr and nCr
- Arrangements with repetition
- Circular permutations
- Word arrangement, rank of a word
- Selection with restrictions, handshake problems

Mensuration

Medium

- Area and perimeter of 2D shapes
- Volume and surface area of cube, cuboid, cylinder, cone, sphere
- Cost of painting, fencing, flooring
- Melting and recasting of solids
- Inscribed and circumscribed figures
- Water flow in pipes and tanks

Ages

Low

- Present, past and future age equations
- Ratio of ages
- Sum and difference of ages
- Age of family members over years

Mixtures & Alligation

Medium

- Alligation rule
- Mixing two or more quantities
- Replacement of liquid (repeated removal)
- Profit in mixing cheap and costly items
- Milk and water problems

Other Common Topics

Low

- Simplification, BODMAS, approximation
- Square roots, cube roots
- Equations (linear and quadratic)
- Series and sequences (AP/GP)
- Clocks, calendars, coordinate basics

02 REASONING ABILITY

Part A · Foundation

20 Qs · 25 min · ~75 sec each · Moderate

Pattern recognition and structured diagrams. Set up tables and diagrams quickly, since puzzle sets carry multiple questions.

Priority is our preparation guide based on how often the topic is practised for NQT: **High** · **Medium** · **Low**. TCS does not publish an official topic weightage.

Number & Alphabet Series — High

- Arithmetic, geometric and mixed series
- Difference and square/cube patterns
- Alternating and two-step series
- Missing and wrong term
- Letter series with positional values
- Alpha-numeric series

Coding – Decoding — High

- Letter shifting and reverse coding
- Number and symbol coding
- Word to number mapping
- Sentence coding
- Conditional coding
- Mixed (alphabet + number) coding

Blood Relations — High

- Direct and indirect relations
- Family tree construction
- Coded blood relations
- Pointing to a photograph
- Generation-based questions
- Gender-unknown cases

Directions & Distances — High

- Cardinal and ordinal directions
- Net displacement (Pythagoras)
- Shadow based directions
- Turns left/right sequences
- Shortest distance after movement

Seating Arrangements — High

- Linear (single and double row)
- Circular (facing inside/outside)
- Square and rectangular tables
- Conditional and uncertain placements
- Floor and box arrangement

Syllogisms — High

- Categorical statements (All, Some, No)
- Venn diagram method
- Possibility and definite conclusions
- Either-or and complementary pairs
- Reverse syllogism

Analogies — Medium

- Word analogies (function, type, part)
- Number analogies
- Letter analogies
- Classification analogies

Odd One Out — Medium

- Word, number and letter classification
- Number pattern odd one
- Alphabet cluster odd one

Clock & Calendar — Medium

- Angle between hands
- Hands overlap, opposite, right angle
- Gain and loss of time
- Day of the week for any date
- Leap year and odd days
- Repeating calendars

Ranking & Ordering — Medium

- Position from left/right/top/bottom
- Total number of persons
- Interchange of positions
- Comparison-based ordering
- Age and height ordering

Alphabet & Number Tests

Medium

- Position and rank in alphabet
- Meaningful words from letters
- Letter-pair removal and counting
- Digits and arrangement counting
- Mirror/reverse alphabet

Symbol Operations & Inequality

Medium

- Redefined operators (@, #, $)
- Coded inequalities
- Mathematical operator swapping
- Conclusion-based inequality

Other Possible Topics

Low

- Data arrangement and puzzles
- Statement and assumption/conclusion
- Cubes and dice (basic)
- Venn diagrams (basic)

03 VERBAL ABILITY

Part A · Foundation (Updated Pattern 2027)

25 + tasks · 25 min · timed per item

Fully restructured. Three task types, each with a hard per-question timer. Work on speed, memory and email structure.

Priority is our preparation guide based on how often the topic is practised for NQT: **High** · **Medium** · **Low**. TCS does not publish an official topic weightage.

Sentence Completion (20 items, 25 s each) — High

- Grammar and tense usage
- Subject-verb agreement
- Articles, prepositions, conjunctions
- Pronouns and modifiers
- Vocabulary: synonyms, antonyms, word choice
- Idioms and phrases, collocations
- Error spotting and correct usage
- Parallelism and sentence structure

Passage Recall (4 + 4 tasks, 30 s read + 90 s write) — High

- Passage disappears after 30 seconds
- Read for key points, names, numbers, sequence
- Reconstruct in your own words from memory
- Keep sentences short and factual
- Practise chunking: who, what, when, why
- Typing speed and accuracy matter

Email Writing (1 task, 9 min) — High

- Subject line, greeting, closing
- Formal tone and clarity
- Structure: purpose, details, action, thanks
- Complaint, request, apology, follow-up, information emails
- Grammar and spelling accuracy
- Word limit and time management

Instruction Screens — Low

- Short timed screens before each task
- Read them quickly, the timer runs
- Know the format in advance via mocks

Skills Tested Overall — Medium

- Grammar accuracy
- Vocabulary range
- Short-term memory
- Professional written communication
- Typing speed

04 ADVANCED QUANTITATIVE + REASONING

Part B · Advanced

14 Qs · 25 min · ~107 sec each · High

Harder, multi-step and set-based questions. Choose carefully, and answer data sets first since they give several questions at once.

Priority is our preparation guide based on how often the topic is practised for NQT: **High** · **Medium** · **Low**. TCS does not publish an official topic weightage.

Data Interpretation — High

- Tables, bar, line, pie charts
- Mixed and caselet DI
- Percentage, ratio, average from data
- Growth, CAGR and comparison
- Approximation techniques

Set Theory & Venn Diagrams — High

- Two and three set problems
- Union, intersection formulas
- Only/exactly/at least counts
- Survey based questions
- Maximum and minimum overlap

Data Sufficiency — High

- Statement I and II sufficiency
- Quantitative and logical forms
- Combined statement analysis
- Yes/No and value type questions

Syllogisms (Advanced) — Medium

- Multiple statements and conclusions
- Possibility cases
- Venn diagram with 3-4 terms

Advanced Probability — Medium

- Conditional probability, Bayes basics
- With/without replacement
- Expectation-style questions
- Geometric and arrangement-based probability

Permutation & Combination (Advanced) — High

- Restricted arrangements
- Distribution of objects (identical/distinct)
- Circular and necklace
- Digits and numbers formation
- Derangement-style problems

Partnership — Medium

- Investment x time ratio
- Entry and exit of partners
- Working and sleeping partners
- Profit sharing with salary

Critical Reasoning — Medium

- Assumption, inference, strengthen/weaken
- Argument evaluation
- Course of action
- Cause and effect

Analytical Puzzles — High

- Multi-variable arrangement puzzles
- Scheduling and ordering puzzles
- Truth/liar and logic puzzles
- Grouping and selection puzzles
- Tournament and ranking puzzles

Advanced Word Problems — Medium

- Time-speed-work combined
- Mixtures with multiple stages
- Sequences and series (AP/GP) application
- Number system with remainders (advanced)

05 ADVANCED CODING

Part B · Advanced

2 problems · 90 min · Easy ~35 min + Medium ~55 min

Languages: C, C++, Java, Python, Perl. Judged on hidden test cases. Always handle edge cases and mind time complexity.

Priority is our preparation guide based on how often the topic is practised for NQT: **High** · **Medium** · **Low**. TCS does not publish an official topic weightage.

Programming Fundamentals — High

- Input/output formats, multiple test cases
- Loops, conditions, nested logic
- Functions and parameter passing
- Data types, overflow, integer division
- Pattern printing (pyramids, diamonds)
- Number printing and series generation

Arrays — High

- Traversal, min/max, second largest
- Rotation and reversal
- Prefix sums and difference arrays
- Kadane's (max subarray)
- Duplicates, missing and repeating elements
- Merging and partitioning
- Rearrangement (positive/negative, zeros to end)
- Frequency counting

Matrix / 2D Arrays — Medium

- Row, column and diagonal traversal
- Spiral and zig-zag traversal
- Rotation and transpose
- Saddle point, boundary sum
- Search in sorted matrix

Strings — High

- Palindrome, anagram, pangram
- Reverse words and sentences
- Character frequency
- Substrings and subsequences
- Compression and run-length encoding
- Pattern matching, longest common prefix
- Case conversion, vowels and consonants
- Parsing and tokenising input

Math & Number Theory — High

- Prime check, sieve of Eratosthenes
- GCD, LCM, Euclid's algorithm
- Factorial, Fibonacci, power
- Perfect, Armstrong, strong numbers
- Modular arithmetic and fast exponentiation
- Base conversion
- Digit manipulation (sum, reverse)
- Combinatorics basics (nCr)

Sorting & Searching — High

- Bubble, selection, insertion sort
- Merge and quick sort
- Binary search and its variants
- Custom comparator sorting
- K-th largest/smallest
- Search in rotated sorted array
- Counting sort basics

Two Pointers & Sliding Window — Medium

- Pair sum in sorted array
- Remove duplicates in place
- Fixed-size window max/sum
- Longest substring without repetition
- Subarray with given sum

Hashing, Sets & Maps — Medium

- Two-sum and variants
- Group anagrams
- First non-repeating character
- Subarray sum equals K
- Frequency-based problems

Recursion & Backtracking

Medium

- Factorial, Fibonacci recursion
- Subsets and permutations
- N-Queens, Sudoku basics
- Tower of Hanoi
- Combination sum
- Maze/path problems

Stack & Queue

Medium

- Balanced parentheses
- Next greater/smaller element
- Expression evaluation (postfix/infix)
- Queue using stacks
- Monotonic stack basics

Linked List

Low

- Reverse, middle element
- Cycle detection
- Merge two sorted lists
- Remove duplicates, delete nth node

Trees & Graphs

Low

- Binary tree traversals
- Height, diameter, level order
- BST insert/search
- BFS and DFS
- Shortest path and connected components

Greedy & Dynamic Programming

High

- Activity selection, minimum coins
- Knapsack (0/1)
- Longest increasing/common subsequence
- Grid paths, climbing stairs
- Partition and subset sum
- Edit distance

Bit Manipulation

Low

- Set bit count, parity
- Power of two check
- XOR tricks (single number)
- Swap and toggle bits

Simulation & Real-world Problems

Medium

- Date, time and calendar logic
- Bank, billing, parking style simulations
- Game and grid movement
- Scheduling and queue processing

Exam-day Coding Strategy

High

- Read constraints, estimate complexity
- Write brute force first, then optimise
- Test with edge cases (empty, one element, max size)
- Handle input format exactly
- Solve Easy fully before Medium
- Partial credit counts: pass as many test cases as possible